



LAB REPORT



COURSE INFORMATION

- ❖ Course Code:
 - ❖ Course Title:
 - ❖ Experiment No:
 - ❖ Experiment Name:
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SUBMITTED TO

- ❖
 - ❖ Department of
 - ❖ East West University
-



SUBMITTED BY

- ❖ Student Name :
- ❖ ID :
- ❖ Section :
- ❖ Department :

East West University

Date of submission:

Data Sheet:

(A) Center of gravity of the compound pendulum =cm

(B) Table for the **time period T** and the **distance** of the **point of suspension** from **end A**.

	Hole No From end A	Distance from end A <i>l</i> cm	Time for 20 oscillations <i>t</i> sec	Mean Time Period <i>T = t / 20</i> Sec
End A	1	5		
	2	10		
	3	...		
	4	...		
	5	...		
	6	...		
	7	...		
	8	40		
	9	45		
End B	11	55		
	12	60		
	13	...		
	14	...		
	15	...		
	16	...		
	17			
	18	90		
	19	95		

Calculation:

Method of measuring length (L):

(From graph)

Length, **AD** = cm.

Length, **BE** = cm.

Mean length, **L** = **(AD+BE) / 2** =cm =m

Corresponding time-period from the graph. (Time corresponding the **ABCDE** line)

T =sec.

$$g = 4\pi^2 \frac{L}{T^2} = = \text{ m/sec}^2$$

Standard value (Theoretical value) of **g** = **9.81 ms⁻²**

$$\text{Percentage Difference} = \frac{\text{Standard value} - \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Percentage difference =

Result:

Acceleration due to gravity, **g** =

Discussions: